FANUC Robotics SYSTEM R-30*i*A Controller Continuous Turn User’s Guide

MARUCCNTR03061E REV D

Version 7.20 and later

About This Manual

This manual can be used with controllers labeled R-30*i*A or R-J3*i*C. If you have a controller labeled R-J3*i*C, you should read R-30*i*A as R-J3*i*C throughout this manual.

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FRA Patent List

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VersaBell, ServoBell and SpeedDock Patents Pending.

Conventions

This manual includes information essential to the safety of personnel, equipment, software, and data. This information is indicated by headings and boxes in the text.

> **Warning**
>
> **Information appearing under WARNING concerns the protection of personnel. It is boxed and in bold type to set it apart from other text.**

Caution

Information appearing under **CAUTION** concerns the protection of equipment, software, and data. It is boxed to set it apart from other text.

Note Information appearing next to **NOTE** concerns related information or useful hints.

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Safety

FANUC Robotics is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. FANUC Robotics therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC Robotic systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

FANUC Robotics therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC Robotics training course and become familiar with the proper operation of the system. Persons responsible for programming the system-including the design, implementation, and debugging of application programs-must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People and Equipment Safe

The safety of people is always of primary importance in any situation. However, equipment must be kept safe, too. When prioritizing how to apply safety to your robotic system, consider the following:

- People
- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and **not** replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC Robotics training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points* . Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and *work envelope* to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC Robotics recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

> **Warning**
>
> **Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.**

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

> **Warning**
>
> **Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.**

  — Using a low motion speed, single step the program for at least one full cycle.
  — Using a low motion speed, test run the program continuously for at least one full cycle.
  — Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that an applicator bell cup can continue to spin at a very high speed even if the robot is idle. Use protective gloves or disable bearing air and turbine air before servicing these items.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Warning

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.

- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

> **Warning**
>
> **Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.**

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

Note Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

Chapter 1

OVERVIEW

Contents

The *continuous turn* option allows a controlled axis to rotate continuously, without limitation.

In normal operation, a robot axis is limited to a restricted number of turns from the zero position. An axis can only turn so far in one direction until the joint limit is reached and the axis must be unwound.

In continuous turn, there are no joint limits or turn limits. An axis can rotate indefinitely in the same direction without limitation. There is no absolute position of an axis beyond one turn. All positions of the continuous turn axis are mapped to the range [-180°, +180°]. For example, if an axis rotates 380 degrees, the position of the axis becomes 20 degrees.

1.1 MODES OF OPERATION

There are two modes of operation possible with the continuous turn option. They are:

- Constant velocity mode, also called the continuous rotation mode (CR).
- Non-continuous rotation mode (NCR).

1.1.1 Continuous Rotation Mode

In the Continuous Rotation mode (CR), the axis rotates continuously at a specified velocity regardless of the motion of the other robot axes or auxiliary axes, except at the start and termination of continuous turn.

1.1.2 Non-Continuous Rotation Mode

In the Non-Continuous Rotation mode (NCR), the continuous turn axis has position range within one turn and is coordinated with the other axes, but has no rotational limits.

When the axis moves, it will take the shortest rotational distance to the desired position.

1.2 FEATURES

1.2.1 Overview

In a standard FANUC Robotics controller without continuous turn, the controlled axes are limited to a finite number of turns before a joint limit is reached. Without continuous turn, an axis cannot remain spinning while the rest of the robot completes a motion.

In a FANUC Robotics robot controller that has the continuous turn option installed, the controlled axes are not limited in the number of turns they can make. A continuous turn axis will never reach a joint limit.

This section contains some of the other features included in the continuous turn option.

1.2.2 Motor

The continuous turn option allows an axis to be treated as a motor rather than as a robot axis.

In the continuous rotation mode of operation, the desired velocity of the continuous turn axis is specified. This provides the capability of driving extended devices that require a motor drive, but also need to be coupled with the robot motion. Possible applications that might need this include conveyor drives, pump drives, grinders, sprayers, and so forth.

1.2.3 Independence

The continuous rotation can be independent of robot motion but can also be synchronized with robot and group motion. The robot can stop while the continuous turn axis keeps spinning.

1.2.4 No Rotational Limits

Another benefit is that the continuous turn axis can be stopped and indexed to different positions, without regard to rotational limits.

You can specify relative motions in which the desired position for the axis can be thousands of turns from the current position.

1.2.5 Shortest Distance

With continuous turn, the axis will always move the shortest rotational distance to the desired position. This reduces cycle time because the axis never needs to "unwind."

1.2.6 Mastering

Continuous turn maintains the axis mastering. The axis position is always correctly retained, even after continuous turn is disabled.

Re-mastering is not necessary even if you have rotated the axis for thousands of turns.

For more information on how to master a robot, refer to the appendix on "Mastering," in the appropriate controller-specific and application-specific *FANUC Robotics Setup and Operations Manual*.

1.3 SOFTWARE COMPATIBILITY

1.3.1 Overview

This section contains information about the compatibility of continuous turn, including possible outcomes when used with other options.

1.3.2 Constant Path

The Constant Path motion functionality is **completely disabled** when the Continuous Turn option is loaded on the controller. Constant Path will remain disabled even though you might not have any continuous turn axes set up.

1.3.3 Multiple Motion Groups

KAREL programming supports only one motion group. When programming in KAREL, only one continuous turn axis is available.

In teach pendant programming, you can have multiple motion groups. However, you cannot specify the continuous turn velocity independently for each motion group. You can only specify the continuous turn velocity for all axes that have continuous turn enabled.

This means that when you execute group motion with continuous rotation, all groups with continuous turn will start to turn simultaneously, at the same velocity.

Also with multiple motion groups, the system variable:$MRR_GRP[i].$SV_OFF_ENB must be FALSE for each axis in all groups. This is a precautionary measure in case the first motion after a COLD START involves continuous turn. In this case there could be a synchronization problem between the motion groups when releasing the brakes and starting continuous rotation.

1.3.4 Other Options

This section contains information about options that might result in unexpected outcomes when used with continuous turn.

DSP/NO DSP

Using the options DSP DRY RUN and NO DSP will destroy mastering data for the continuous turn axis if continuous turn is enabled. You should disable continuous turn if you want to use either DSP DRY RUN or NO DSP.

Can Be Used With

At this release, continuous turn has been qualified for the following options:

- Single and multiple motion group teach pendant programs without application programs
- All KAREL programs
- TurboMove

Chapter 2

HARDWARE AND SOFTWARE

Contents

2.1 HARDWARE

Operating the Continuous Turn option requires the following hardware equipment:

- Robot
- Controller
- Memory
- Coprocessor
- Peripheral devices

Robot

Continuous turn can run on the last axis of any FANUC Robotics robot model except for door openers, hood/deck openers, C-100, or any other robot that mechanically can not support unlimited rotation of the last axis, or which exceeds the gear ratio.

Refer to Appendix A for more information about gear ratios and supported robot models.

Extended axes are supported depending on their gear ratio and mechanical design.

Memory

Continuous turn requires 30 KB of space in CMOS memory.

Coprocessor

Continuous turn works independently of coprocessor speed.

Peripheral Devices

Continuous turn requires no additional peripheral devices but was designed for use with extended axes.

Note The only hardware limitations for continuous turn are the external cables and wires attached to the robot or extended axis. When designing your workcell to include continuous turn, make sure you consider the possibility of any external cables.

2.2 SOFTWARE

Continuous turn is an option for the core motion system. The continuous turn software is supplied on a separate memory card.

Continuous turn is loaded as an option after the robot controller system has been loaded. For a detailed description of how to load the continuous turn option, refer to the chapter "Software Installation" in the controller-specific *FANUC Robotics Software Installation Manual.*

Continuous turn can also be loaded as an update to existing system software.

Note The Constant Path Option will be automatically disabled when the Continuous Turn option is installed.

Chapter 3

GENERAL SETUP

Contents

3.1 ENABLING CONTINUOUS TURN

This section contains the procedure for enabling continuous turn.

There are three system variables that control the use of continuous turn. You can modify the values of these system variables from the continuous turn setup screen.

After loading continuous turn, you must modify the system variables for each motion group, in order to enable the option.

Use Procedure 3-1 to enable continuous turn by modifying the proper system variables.

Table 3–1. SETUP Continuous Turn Screen Items

ITEM	DESCRIPTION
Group	This item indicates the number of the group.
Continuous Turn Axis Num	This item indicates which axis in the group has continuous turn enabled. Zero means continuous turn is completely disabled for the specified group.
Numerator of Gear Ratio	This item indicates the exact integer value of the gear ratio numerator, for the continuous turn axis.
Denominator of Gear Ratio	This item indicates the exact integer value of the denominator of the gear ratio for the continuous turn axis.

Refer to the *Software Reference Manual* for further information regarding continuous turn system variables.

Procedure 3-1 Enabling Continuous Turn

Conditions

- The continuous turn option has been installed. Refer to the appropriate controller-specific *FANUC Robotics Software Installation Manual* for more information.

Steps

1. Select SETUP from the teach pendant main menu.
2. **If no other options are loaded,** you will be presented with the continuous turn setup screen. Go to Step 3 .

   OR

   If other options are loaded,

   a. Press F1, [TYPE].

   b. Select Cont. Turn. You will see a screen similar to the following.

```
SETUP Continuous Turn      G1     JOINT 10%
1: Group: 1
2: Continuous Turn Axis Num:   0
3: Numerator of Gear Ratio:   0
4: Denominator of Gear Ratio  0
```

Warning

Never change continuous turn system variables outside of the setup menu. Bypassing the setup menu can corrupt the controller.

3. Select Continuous Turn Axis Num:

   This field indicates which axis in the group has continuous turn enabled. Zero means continuous turn is completely disabled for the specified group.

4. Enter the robot axis number that you want to have continuous turn. **Only the last axis of a robot or an extended axes can have continuous turn.**

   After you type a continuous turn axis number, the value of $PARAM_GROUP[i].$contaxisnum is set.

5. Select Numerator of Gear Ratio:

   This field indicates the exact integer value of the gear ratio numerator, for the continuous turn axis.

   The gear ratio for the continuous turn axis is specified by:

   cn_gear_n1/cn_gear_n2

   where

   - typically *cn_gear_n1>cn_gear_n2.*
   - *cn_ gear_n1* gives the number of motor revolutions per *cn_gear_n2* revolutions of the joint.
   - The maximum value is 32766.
   - gear ratios must be less than 4000 *(cn_gear_n1/cn_gear_n2<4000).*

   This sets the value of $PARAM_GROUP[i].$cn_gear_n1.

6. Select Denominator of the Gear Ratio:

This field indicates the exact integer value of the denominator of the gear ratio for the continuous turn axis.

The gear ratio for the continuous turn axis is specified by:

cn_gear_n1/cn_gear_n2

where

- typically *cn_gear_n1>cn_gear_n2.*

- The maximum value is 32766.

  This sets the value of $PARAM_GROUP[i].$cn_gear_n2.

7. When you have finished changing the system variable values, press F4, DONE. The changes will then automatically take effect.
8. You must perform a Cold start to complete setup. A message will be displayed telling you when to do so.

You have finished modifying the system variables necessary to setup up continuous turn. Now proceed to Chapter 4 *OPERATIONAL RULES* for rules on how to operate continuous turn.

3.2 MASTERING

The continuous turn option requires the exact gear ratio information in order to maintain mastering positions over all rotations.

Additional mastering information is maintained during motion on the continuous turn axis. This additional information is kept in the "read-only" $DMR_GRP[i].$shift_error system variable.

3.2.1 Loss of Mastering Data

If you change any of the three system variable setup values during system setup, the additional mastering information, mentioned above, is cleared out. This means you could potentially lose mastering accuracy if the system variable setup values are changed after the initial continuous turn installation.

Even though the accuracy loss is less than one encoder pulse count, if you change the continuous turn setup frequently, the loss can accumulate.

Warning

There is the potential of losing mastering data if you change the setup values for continuous turn frequently.

If accuracy is lost, re-mastering might be necessary.

For more information about how to master a robot, refer to the appendix on "Mastering," in the appropriate application-specific *FANUC Robotics Setup and Operations Manual*.

Chapter 4

OPERATIONAL RULES

Contents

4.1 AXES/STARTUP

The following axes and startup related rules must be applied when you operate continuous turn.

4.1.1 Only One Continuous Turn Axis

You are allowed one continuous turn axis per motion group. This implies that there can be up to three continuous turn axes on one controller; one per group.

When the motion group involves a robot with kinematics, only the last axis of the robot can be used as a continuous turn axis, provided that the mechanics of the robot permits it.

4.1.2 Auxiliary and Extended Axes

When the motion group involves auxiliary axes, any of the axes can be designated as the continuous turn axis.

If there are extended axes that are integrated, only the last extended axis can be used as a continuous turn axis.

If the axes are non-integrated, any of the extended axes can be used provided there are no hardware limitations.

4.1.3 Configuration

The positional range of a continuous turn axis must be +/-180°.

Changing an axis to or from continuous turn can only be done on a Cold start.

When removing continuous turn from an axis, it is possible that the axis can be far outside the joint limits. You must first jog the axis back within the joint limits. The controller will only allow motion that returns the joint within the limits.

4.1.4 Gear Ratio

The maximum gear ratio allowed is 4000:1.

The maximum value of the numerator or denominator of the gear ration is 32766.

Refer to Appendix A for more information about gear ratios.

4.1.5 System Variable Initialization

When you use the robot axis for continuous rotation, the setting of $UTOOL should be such that there is no x or y component in the tool frame. Only z offsets are allowed.

Any other setting indicates that linear and circular motion will not be maintained when going from continuous rotation to non-continuous rotation.

If you set an incorrect or invalid continuous turn axis number, a warning message will be produced at Cold start. The Motion Reset Record variable will be set to zero. The user variable in the parameter group will remain unchanged.

In multiple groups, the $SV_OFF_ENBL system variable must be false for all groups. For example, brakes in groups that are coordinated through KAREL or teach pendant programming, must not engage.

4.1.6 Mastering Data

With continuous turn installed and running, the mastering data for the continuous turn axis is automatically changed.

After continuous turn is installed, all off-line stored or written mastering data for the continuous turn axis is no longer valid. The robot will lose mastering on the continuous turn axis if these old numbers are re-install ed. This includes saved system variables.

4.2 STARTING AND TERMINATING CONTINUOUS TURN

The following operational rules must be applied when you start or terminate continuous turn motion.

4.2.1 KAREL Program

In KAREL, continuous rotation will start at the first motion after the continuous rotation command.

In KAREL, continuous rotation will terminate at the beginning of the next motion segment, after the termination command.

All axes in the group, including the continuous turn axis, will stop at the beginning of the motion that ends continuous rotation. The continuous turn axis will start to decelerate only after the other axes have come to a complete stop.

All continuous motion types of the previous motion will be ignored upon ending continuous rotation. For example: CONT100, VARDECEL, and so forth, will be ignored, and all axes in the group must stop.

Refer to Chapter 5 *PROGRAMMING* , for more information about continuous turn motions in a KAREL program.

4.2.2 Acceleration

During continuous rotation mode, the acceleration of the continuous turn axis is specified by the following system variables:

- $PARAM_GROU P[i].$ACCEL_TIME1[contaxisnum]
- $PARAM_GROU P[i].$ACCEL_TIME2[contaxisnum]

The acceleration time will not change until the motion following continuous rotation.

During non-continuous rotation mode, the acceleration of the continuous turn axis is determined by the system variables $ACCEL_TIME1 and $ACCEL_TIME2 for joint motions, and $CART_ACCEL1 and $CART_ACCEL 2 for Cartesian motions.

These accelerations can be changed by the normal operation of the motion system, to speed up short motions. The short-motion speed-up in acceleration can be changed via $ACCEL_OVRD and $SHORTMO_MGN during joint motions.

Acceleration Switch

A switch, $CN_USR_GRP.$cn_grp_acc, is provided during continuous rotation mode, to ensure that all the axes in the group will have the same acceleration time as the continuous turn axis.

This switch can be changed only at Cold start.

Refer to the *Software Reference Manual* for further information on the $CN_USR_GRP.$cn_grp_acc system variable.

4.2.3 Deceleration

When you end continuous rotation, the continuous turn axis might rotate more than one turn as it smoothly decelerates to a stop. This depends on the acceleration times for the joint axis.

Speed

Robot motion might slow down during the ending motion, if the continuous turn axis becomes the speed limiting axis.

Synchronization

Synchronization with other groups and other axes in the same group, might be lost during the motion that ends continuous rotation.

4.2.4 Moving to the Desired Position

When the robot goes to the desired position upon ending continuous rotation, the speed of the continuous turn axis will be calculated as the ratio of the robot motion speed over the maximum speedlim or speedjntlim, depending on whether the motion was linear, circular, or joint.

When ending continuous rotation, the continuous turn axis will move to the desired position in the same direction as the continuous rotation. It will not "spin backwards." This feature is controlled by the system variable $CN_SAME _DIR, and can be disabled.

Refer to the *Software Reference Manual* for further information on the $CN_USR_GRP.$cn_same_dir system variable.

4.2.5 Ending Motion

The continuous turn option must be ended only on certain motions.

Joint Motions

Continuous rotation must be ended on a joint motion if the continuous turn axis is the last axis of the robot. It can be ended on either a joint, linear, or circular motion if it is an extended axis.

Relative Motions

Continuous rotation cannot be ended on a relative motion. This holds for teach pendant programs and KAREL programs. The program is paused and an error message is posted.

Process Motions

Continuous rotation cannot be ended on process motions. For example, continuous rotation cannot end on an Arc Start motion.

4.2.6 Recording Points

All points recorded while the continuous turn feature is enabled will be recorded in turn 0.

During playback of a previously recorded point that was not in turn 0, the desired position will be automatically wrapped into turn 0.

4.3 SPEED LIMITS

The following speed related rules must be applied when you operate continuous turn.

During continuous rotation, the rotational velocity is specified as a percentage of the maximum rotational velocity of the joint. This velocity is subject to teach pendant speed overrides and program overrides.

4.3.1 Limited by Mode of Operation

The maximum rotation velocity of a continuous turn axis is limited depending on the mode of operation.

Continuous Rotation: Limited by Maximum Axis Velocity (function of max motor speed and gear ratio).

Non-Continuous: Limited to 180 ° per INTR loop. The formula is:

$$max\_joint\_vel = 180(deg) * \frac{1}{ITP\_TIME * RATE(ms)} * 1000(ms/sec)$$

- where *RATE* = maximum of $JOINT_RATE, $LINEAR_RATE, or $CIRC_RATE.

Table 4–1 for *ITP_T IME* = 28, shows the speed limit in RPM and degrees/second:

Table 4–1. NonContinuous Turn Maximum Joint Axis Speed

RATE	RPM	DEGREE/SECONDS
1	1071	6429
2	535	3215
3	357	2143

All motion will stop and an error message will be issued if the program attempts to exceed the maximum speed during non-continuous rotation.

4.3.2 $CN_TURN_NO

$CN_TURN_NO is a system variable which shows the integral number of turns spun by an axis, since the start of continuous rotation. This variable is only valid for speeds under:

$$Valid\ spd\ lim = 180(deg) * \frac{1}{ITP\_TIME * RATE(ms)} * 1000(ms/sec)$$

This translates to a maximum 1071 RPM for the joint. A warning message will be issued if the motion, without overrides, can exceed this speed.

Higher speeds are obtainable for the axis, but $CN_TURN_NO is not valid at these higher speeds.

Refer to the *Software Reference Manual* for more information about the $CN_TURN_NO system variable.

4.4 MOTION

The following motion rules must be followed when you operate the continuous turn option.

4.4.1 FWD/BWD Single Step Motions

The system variable $CN_USR_GRP.$cn_step_enb acts as a switch by controlling CTV (continuous) motion when you single step through a program. You can change the value of this variable either on the teach pendant System Variables screen or in a KAREL program.

- When $CN_USR_GRP.$cn_step_enb = FALSE the continuous turn axis (in forward and backward single step motion), will not spin but will move the shortest rotational distance to the taught position.
- When $CN_USR_GRP.$cn_step_enb = TRUE the continuous turn axis does not move when in forward and backward single step mode.

Refer to Table 4–2 for a description of how the system variable $CN_USR_GRP.$cn_step_enb affects the motion of the continuous turn axis (while in STEP mode).

Note After a PAUSE or SHIFT-RELEASE, pressing SHIFT and BWD causes the continuos turn axis to stay where it is for CTV motion.

Table 4–2. $CN_USR_GRP.$cn_step_enb System Variable

Value of $CN_USR_GRP.$cn_step_enb	Single Step Mode	FWD/BWD	Continuous Turn Axis
FALSE (moves to taught position when in STEP)	-	BWD	moves to taught position
	TRUE (ON)	FWD	moves to taught position
	FALSE (OFF)	FWD	rotates continuously
TRUE (does not move when in STEP)	-	BWD	stays at start position
	TRUE (ON)	FWD	stays at start position
	FALSE (OFF)	FWD	rotates continuously

Note The continuous turn system variable $CN_SAME_DIR, has no effect during forward (FWD) and backward (BWD) stepping.

4.4.2 Shortest Rotational Distance

When you move the robot to positions in non-continu ous rotation mode, the continuous turn axis will move the shortest rotational distance.

Note If the continuous turn axis is the last axis of the robot, and the motion type is linear or circular, the robot motion type (linear/circular) takes precedence over the non-continuous rotation motion.

For example, in a KAREL program, if the $MOTYPE is linear(7), a MOVE AXIS BY command will be treated as if the axis is NOT a continuous turn axis. Similar results hold for REL motions in teach pendant programs.

4.4.3 Jog

With continuous turn enabled, the selected axis can be jogged infinitely in either direction. The current position of the axis is displayed on either the CRT or teach pendant while jogging. The displayed position will wrap from +180 to -180 each time the axis crosses this boundary.

The continuous turn axis will move in non-continuous rotation mode during jogging.

In non-continuous rotation mode, the continuous turn axis can be jogged past 180°. The jog speed override is used when jogging.

The continuous turn axis will enter continuous rotation, if $CONTAXISVEL is set, when the robot is moved with the teach pendant under the KAREL DATA page, using the MOVE_LN and MOVE_JT buttons.

4.4.4 Hold

When a HOLD is executed from within a KAREL program, under continuous rotation mode, the continuous turn axis will not stop. Program UNHOLD will have no effect on the continuous turn axis.

When a CANCEL is executed from within a KAREL program, the continuous turn axis will stop but will start turning at the beginning of the next motion, unless $CONTAXISVEL was set to zero before that motion.

When a STOP is executed from within a KAREL program, the continuous turn axis will stop turning until a RESUME is issued.

4.4.5 Resume

On an operator panel HOLD or a teach pendant HOLD, the continuous turn axis will stop turning. It will resume turning when CYCLE START or SHIFT FWD are pressed only if the other axes in the group have a motion segment to resume.

If all other axes had completed their motion before HOLD was pressed, the continuous turn axis would not start turning until the next motion command.

4.4.6 Positive Direction Bias

When you move the robot in the non-cont inuous rotation mode, 180°motions are biased towards rotation in the positive direction.

There is a small region of degrees, around 180°, where motions are biased to move in the positive direction. If the shortest rotational distance is greater then -179.99926°, then the axis will rotate in the positive direction rather than the negative direction.

For example, if the commanded rotation is 180.0°, the axis will rotate in the positive direction. If the commanded rotation is 180.00074°, the axis will still rotate in the positive direction, even though the shortest rotational distance is -179.99926°.

The bias is positive when moving forward (FWD) or backward (BWD) along KAREL paths. The bias is reversed when using BWD and single stepping the program.

Chapter 5

PROGRAMMING

Contents

5.1 OVERVIEW

You can program continuous turn either through KAREL or the teach pendant editor. There are two modes of operation for continuous turn: continuous rotation, and non-continuous.

Continuous

When operating in continuous rotation mode (CR), you must program the desired rotational velocity.

- In the teach pendant editor, you select a motion and attach an optional field for the continuous turn velocity. Refer to Section 5.2 for more information about programming continuous turn with the teach pendant editor.
- In KAREL, you specify the desired continuous turn rotational velocity by setting a group motion variable before motion. Refer to Section 5.3 for more information about programming continuous turn in KAREL.

Non-Continuous

When operating in non-continuous rotation mode (NCR), there are no special programming statements or commands to use the unrestricted joint motion feature of continuous turn.

Refer to Section 5.4 for information on how the execution of a program differs with and without the continuous turn option enabled.

5.2 TEACH PENDANT PROGRAMMING

This section discusses how to program continuous turn using the teach pendant editor.

The continuous rotation feature is located in the motion option page. You can add the continuous turn velocity option at the end of a motion line.

The default motion line is for no continuous rotation.

Procedure 5-1 Adding Continuous Turn to a Motion Instruction

Conditions

- All personnel and unnecessary equipment are out of the workcell.
- You want to add continuous turn velocity to the end of a motion instruction.
- The program has been created and all detail information has been set correctly. If not, refer to the chapter on "Planning and Creating a Program," in the application-specific *FANUC Robotics Setup and Operations Manual.*

Steps

1. Press SELECT.
2. Display the appropriate list of programs:
   a. Press F1, [TYPE].
   b. Select the list you want:
      - All displays all programs.
      - TP Programs displays all teach pendant programs.
      - KAREL Progs displays all KAREL programs.
      - Macro displays all macro programs.
3. Select the name of the program you want to modify.
4. Press EDIT.
5. Press and hold down the DEADMAN switch, while turning the teach pendant ON/OFF switch to ON.
6. Move the cursor to the end of the instruction for the line number of the motion instruction you want to modify. See the following screen for an example.

```
JOINT 10%
1: J  P[1] 100% FINE
```

7. Press F4, [CHOICE]. You will see the following screen.

```
Motion Options
1:  No option     5:  Offset, Pr[]
2:  ACC           6:  Incremental
3:  Skip, LBL[]   7: CTV
4:  Offset        8:
                                 JOINT 10%
1: J  P[1] 100% FINE
```

8. Select CTV. You will see a screen similar to the following.

```
JOINT 10%
  1: J  P[1] 100% FINE CTV
```

9. Enter a CTV option. Valid values range from -100 to +100. See the following screen for an example.

```
JOINT 10%
1:  J P[1] 100% FINE CTV 50
```

Note CTV0 means that the continuous turn axis does not rotate. However, CTV0 is treated as continuous rotation and must be ended as all other continuous rotation motions. When ending CTV0, the continuous turn axis will always take the shortest distance to the desired position.

When this option is added, continuous rotation at the specified velocity is turned on for the instruction. If there is no option, continuous rotation will be turned off automatically.

Note When you are in single STEP mode, CTV motions operate differently depending on the value of the system variable $CN_USR_GRP.$cn_step_enb. For more information on this system variable, refer to Section Section 4.4.1 .

Continuous Termination Type

Continuous termination type is ignored on motions preceding the ending of continuous rotation. Refer to Ignore Continuous Termination Type Motions for an example.

Ignore Continuous Termination Type Motions

```
L P[1] 100mm/sec CONT100 CTV100
--Robot will
stop at end of this
 L P[2] 100mm/sec FINE
--motion, CONT100
is ignored.

L P[1] 100mm/sec CONT100 CTV100
       --Robot will NOT stop since next
 L P[2] 100mm/sec FINE CTV0
--motion, is a
continuous turn
 L P[3] 100mm/sec FINE
--motion with
```

```
0 velocity.
```

5.3 KAREL PROGRAMMING

The system variable $CONTAXISVEL allows you to use continuous turn in two different modes.

1. When the value in $CONTAXISVEL is zero, the continuous turn axis is in the non-continuous rotation mode.
2. When the value in $CONTAXISVEL is not zero, the continuous turn axis is in continuous rotation mode. The value of $CONTAXISVEL is used as the velocity of the continuous turn axis, expressed as a signed percentage of the maximum speed. The direction of rotation for the continuous turn axis is determined by the sign of $CONTAXISVEL (positive/negative).

5.3.1 Non-Continuous Rotation Mode

In non-continuous turn mode:

- KAREL MOVE commands operate with the last axis or auxiliary axis position wrapped to +/- 180°. The joint can be jogged indefinitely in either direction and the displayed joint position will remain within the +/- 180°bounds.
- All motion commands will be effectively restricted to this region. All rotatio ns of the last axis will take the shortest rotational distance to complete the motion.
- The exception to this is the command MOVE AXIS. This is a relative command and can be applied to the continuous turn axis when in the non-continuous rotation mode.

With the MOVE AXIS command, you can rotate the last axis over a very large number of degrees. However, extremely large motions could be less accurate. For example, if the command is MOVE AXIS 6 BY 3600000; this should produce 10000 rotations of the last axis and end with the axis in the same position as it began.

Due to floating point accuracy, the axis might not end exactly at the initial position. If you want to have the axis end exactly at the initial position, use continuous rotation. Refer to Section 5.3.2 for more information.

5.3.2 Continuous Rotation Mode

In continuous rotation mode:

- Setting $CONTAXISVEL to a value other than zero specifies that you want continuous rotation mode for an axis. Axis motion begins when you issue the next MOVE statement. See Terminating Continuous Turn I .
- The continuous turn axis stops rotating at the beginning of the next MOVE command, after $CONTAXISVEL has been set to zero. As illustrated by the example in Terminating Continuous Turn I , the continuous rotation mode is terminated by moving to a point after setting $CONTAXISVEL to zero. In this case the motion will not follow the shortest rotational distance.

Terminating Continuous Turn I

```
begin
  $group[1].$CONTAXISVEL=50        --50% of full axis speed
  MOVE ALONG path                  --Axis will start turning
                                   --Continuous axis should rotate
                                   --continuously even after robot
                                   --is done with the path.
  $group[1].$CONTAXISVEL=0         --To stop continuous axis rotation
  $MOVE TO point2                  --Axis stop rotating at the
                                   --beginning of the motion to
                                   --this point. Rotational velocity
                                   --will change.  The end position
  endsample_prog                   --is given by point2
```

- When terminating continuous rotation mode, the continuous turn axis will always turn in the same direction until it reaches the goal position.
- The axis will not "back spin" to reach the goal position. Additionally, the velocity in the termination motion will NOT be the velocity of the previous continuous rotation. That velocity was terminated at the beginning of the termination motion. See Terminating Continuous Turn II .

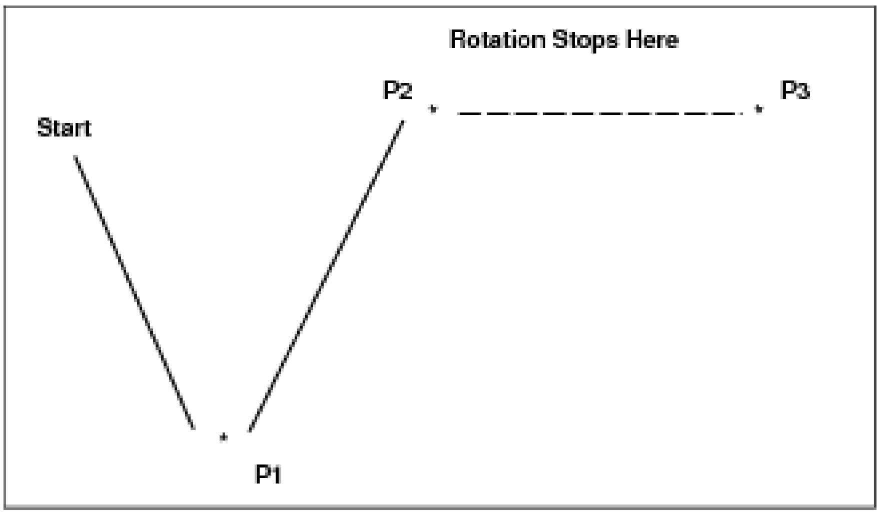
Terminating Continuous Turn II

```
$GROUP[1].$CONTAXISVEL=50
MOVE TO P1
MOVE TO P2
$GROUP[1].$CONTAXISVEL=0
DELAY 5000
MOVE TO P3
```

The code in Terminating Continuous Turn II will do the following:

1. The continuous turn axis will start spinning immediately as the robot starts moving to P1.
2. It will continue to spin as the arm moves to P2 and during the delay of 5 seconds.
3. At the start of the motion to P3, the continuous turn axis will change velocity and stop continuous rotation. Figure 5–1 illustrates this point.

Figure 5–1. Stopping Continuous Rotation on the Final Move



Turn Count System Variable

In continuous rotation mode, the KAREL program has access to the number of turns since the start of the continuous turn motion. This can be used to stop a motion or to trigger some condition.

The softpart system variable $CN_TURN_NO contains the number of turns since the start of the continuous turn motion. Upon initialization, this value is zero. Whenever the continuous rotation motion begins, the value is reset to zero. During continuous rotation motion, the $CN_TURN_NO variable will be incremented for each full turn from the starting position.

In order to gain access to the value of $CN_TURN_NO, the cndef environment must be included in the environment declaration of the KAREL program. Refer to Gaining Access to the Turn Count System Variable for an example.

Gaining Access to the Turn Count System Variable

```
PROGRAM prg_name
%ENVIRONMENT cndef
```

```
begin
$group[1].$contaxisvel=50
move to p1                           --Rotation starts here, $cn_turn_no
--or                                 --is reset automatically to zero
$group[1].$contaxisvel=0
move to p1                           --Rotation stops here
--or
if $cn_usr_grp[1].$cn_turn_no > 100 then
                                     --Perform operation based on
                                     --number of turns.
endif
end prg_name
```

Using the example in Gaining Access to the Turn Count System Variable , you can turn the axis a specified number of turns and produce better results than with a very large MOVE AXIS command.

Note The value of the system variable $CN_TURN_NO is limited by the storage of an unsigned integer, which is 2,147,483,647 turns on the controller. This number will probably not be reached. However if it is, $CN_TURN_NO will reset back to zero with no error or warning.

Local Condition Handler

Instead of using $CN_TURN_NO you can accomplish the same thing with a separate condition handler that monitors the value of $CN_TURN_NO and stops the motion when it reaches a given value.

Continuous rotation will end on a local condition handler cancel. The example in Using a Local Condition Handler to Monitor $CN_TURN_NO illustrates this point.

Using a Local Condition Handler to Monitor $CN_TURN_NO

```
Portions of KAREL Code
$contaxisvel=50
move to p1              --Rotation starts here.
--or
move to p2,
until DIN[1]=ON         --Rotation stops when this motion is canceled.
endmove
delay(3000)             --No rotation during this delay if previous
                        --motion was cancelled due to local condition.
--or
$contaxisvel=0.0
move to p10             --Rotation normally ends here.
```

$CONTAXISVEL

You might want to leave the continuous turn axis in the continuous rotation (CR) mode, with zero rotational velocity. This can be accomplished by setting $CONTAXISVEL to a very small floating point number, for example 0.000001. This will stop the continuous turn axis but will leave the motion in the continuous rotation mode.

If the absolute value of $CONTAXISVEL goes below EPSILON (0.00001), the continuous turn axis will always take the shortest distance to the goal position, regardless of the sign of $CONTAXISVEL.

Relative & Path Motions

In KAREL, continuous rotation cannot be ended on a RELATIVE motion (MOVE AXIS BY, MOVE ABOUT, and so forth) or on a PATH motion. The motion would be stopped and an error message posted.

5.4 PROGRAM EXECUTION

When continuous turn is on an axis, that axis will only display and move to a range of +/-180° degrees. All motion will be of the shortest rotational distance and will wrap around 180° degrees.

5.4.1 How the Program Executes Differently With Continuous Turn

If a program was written when continuous turn was not enabled, the continuous turn axis might not follow the original taught path. It might now choose a different direction on the continuous turn axis.

In KAREL, the MOVE AXIS BY command will perform differently. See Difference in Execution With Continuous Turn Enabled for an example of how the execution would differ with continuous turn active.

Difference in Execution With Continuous Turn Enabled

```
MOVE TO p0
MOVE AXIS 6 BY 300
MOVE TO p0
```

Without Continuous Turn

Before the installation of continuous turn on the 6th axis, the axis would move to p0, rotate 300°, and then rotate backward 300°to p0 again.

With Continuous Turn

After the installation of continuous turn on the 6th axis, the axis will move the shortest distance to p0, rotate 300°, then move 60°to p0 (the shortest rotational distance).

5.4.2 UTOOL Limitation

If the J6 axis of a 6–axis robot is configured as a continuous turn axis, then the UTOOL for that 6–axis robot should be set to (0, 0, 0, 0, 0, 0). If this restriction is not observed, then Continuous Mode motion might not resume properly after a HOLD or E-stop is executed.

5.4.3 Motion Type Limitation

If the continuous turn axis is the last axis of the robot, and the motion type is linear or circular, MOVE AXIS BY will perform in the normal motion manner.

Chapter 6

TROUBLESHOOTING

Contents

6.1 POTENTIAL PROBLEMS AND SOLUTIONS

Table 6–1 contains descriptions of potential problems you might encounter when you use continuous turn, and their solutions.

Table 6–1. Continuous Turn Common Problems

SYMPTOM	CAUSE	SOLUTION
You are experiencing motor speed limit errors, when using the last axis of the robot under continuous rotation at 100% joint velocity.	This is due to the joint interactions where other robot joints can change the motor speed of the last axis due to gear system interactions.	This problem can be avoided by lowering the continuous rotation velocity.
The controller crashes.	You have changed the system variables for the continuous turn gear ratio and axis number, outside of the setup menu. The setup menu contains protection features that limit the range of numbers that can be entered for the continuous turn axis.	Make sure that you always setup continuous turn via the SETUP menu.
The robot loses mastering data.	With continuous turn installed and running, the mastering data for the continuous turn axis is automatically changed. After continuous turn has been installed, all off-line stored or written mastering data for the continuous turn axis is no longer valid. The robot will lose mastering on the continuous turn axis, if these old numbers are re-installed. This includes saved system variables.	Mastering data is consistent within the robot. You should not record or save old mastering data with continuous turn installed. If you want to reload system variables from a floppy or RAMDISK, do the following to ensure that mastering is not lost on the continuous turn axis:1. Write down the mastering data for the continuous turn axis.2. Record the value of$DMR_GRP[i].$shift_error. 3. Load the system variables.4. Restore the mastering data.5. Shift_error to the previous values.

Table 6–1. Continuous Turn Common Problems (Cont'd)

SYMPTOM	CAUSE	SOLUTION
The continuous turn axis is rotating without the explicit use of a $CONTAXISVEL = <value> statement in the KAREL program.	If the KAREL program contains the %NOLOCKGROUP directive, the motions in the program will include the old value of the system variable $GROUP.$contaxisvel. The reason is that %NOLOCKGROUP prevents the KAREL translator from clearing the $GROUP[i].$contaxisvel value from the previous program. If the previous program ended/aborted in the middle of continuous rotation, the $CONTAXISVEL would still have a value. This would mean that the axis would start turning again on the first motion of the new program.	You should avoid the use of %NOLOCKGROUP. However, if you must use it, set $CONTAXISVEL=0.0 at the beginning of the KAREL program.
The direction of rotation for the continuous turn axis changes, during a 180°motion.	For exactly 180°motions, the direction of rotation of the continuous turn axis might change depending on load, speed, stopping conditions, and single-step operations. This means that the axis might rotate either in the positive or negative direction when moving to the goal position. The reason is that continuous turn axis will choose the shortest distance less than 180°. At a motion that is exactly 180°, both positive and negative rotations produce the same distance. When the robot stops moving, it will have a slight error of much less than a degree on either side of the stopping position. This small error will bias the choice of rotation toward positive or negative rotation.	Avoid 180°motions if possible, by changing the value to be distinctly greater or less than 180°.

Table 6–1. Continuous Turn Common Problems (Cont'd)

SYMPTOM	CAUSE	SOLUTION
You receive error code CNTR-010 Option Not Allowed	This error can occur even when the motion does not use the independent extended axis option.	Make sure the system variable $GROUP[i].$ext_indep is set to FALSE. If this variable is accidentally set to TRUE, continuous turn will not function.
You receive error codeMOTN-018 Position Unreachable	This error occurs when you end continuous rotation on the last robot axis. If you single-step through the program, an error will not occur. However when you run the program, an error will occur.	Continuous turn axis is the last axis of the robot. Change the motion type (motype) to JOINT and teach an extra point, if necessary, to end continuous rotation.

Appendix A

GEAR INFORMATION

Contents

A.1 GEAR INFORMATION FOR ROBOT MODELS

This appendix contains the gear information that you need to run continuous turn, as it relates to each FANUC Robotics robot model that supports the option.

Table A–1. Gear Ratio Information for FANUC Robotics Robots

Controller Type	Robot Model	CN_GEAR_N1	CN_GEAR_N2	Order Number
R-30*i*A	R-2000iB/125L	90	1	h600
	R-2000iB/165F	1431	14	h738
	R-2000iB/210F	5252	43	h601
	M-710i/New	2006	23	h641
	M-710i/ShortArm	2006	23	h640
	Arc Positioner (500kg) - J2	1911	17	h871
	One axis positioner (300kg)	2793	16	h879
	One Axis Positioner (1000kg)	2520	12	h880
R-J3*i*B	ArcMate 100 iBe	50	1	h778
	ArcMate 100 iB S/A	50	1	h728
	ArcMate 100 iBe S/A	50	1	h729
	ArcMate 120 iB S/A	50	1	h726
	ArcMate 120 iB/10L S/A	50	1	h656
	ArcMate 120 iBe	50	1	h779
	ArcMate 120 iBe S/A	50	1	h725
	LRMate200iB/3L(B262)	50	1	h767
	LRMate200iB/3L(B264)	50	1	h633
	M-6iB/2HS	30	1	h733
	M-6iB/2HS S/A	30		
	M-6iB/6S, AM100iB/6S	18	1	h662
	M-6iB/6S, AM100iB/6S	50	1	h773
	M-6iB/6T	50	1	h734
	M-6iB/ARCMate100iB	50	1	h770

Table A–1. Gear Ratio Information for FANUC Robotics Robots (Cont'd)

R-J3iB cont'd	M-16iB/10L	50	1	h772
	M-16iB/10LT	50	1	h776
	M-16iB/20 AM120iB/20	50	1	h771
	M-16iB/20T	50	1	h775
	M-410iB/160	101235	1342	h791
	M-410iB/300	113145	976	h792
	M-410iB/450	1421	12	h790
	M-420iA/HS	31	1	h872
	M-421iA	81	1	h877
	M-710iB/45	387	8	h780
	M-710iB/45E	387	8	h781
	M-710iB45T	387	8	h783
	M-710iB/45T-HS	387	8	h653
	M-710iB/70	5345	76	782
	M-710iB/70T	5345	76	784
	M-710iB/70T-HS	5345	76	h654
	M-900iA/260L	5151	44	h789
	M-900iA/350	561	4	h788
	M-900iA/600	243	2	h795
	R-2000i/130U	6413	42	h759
	R-2000iA/100P	1111	8	h798
	R-2000iA/125L	627	8	h744
	R-2000iA/165F	891	8	h740
	R-2000iA/165R	891	8	h742
	R-2000iA/200F/STUD	6413	42	h746
	R-2000iA/200FO	6413	42	h756

Table A–1. Gear Ratio Information for FANUC Robotics Robots (Cont'd)

R-J3*i*B cont'd	R-2000iA/200FO/STUD	6413	42	h758
	R-2000iA/200R	6413	42	h743
	R-2000iA/200R/STUD	6413	42	h757
	R-2000iA/200T-HS	6413	42	h747
	R-2000iA/210F	1111	8	h760
	S-900iB/200	5656	48	h787
	S-900iB/220L	6776	48	h786
	S-900iB/400	6776	48	h785

Glossary

A

abort

Abnormal termination of a computer program caused by hardware or software malfunction or operator cancellation.

absolute pulse code system

A positional information system for servomotors that relies on battery-backed RAM to store encoder pulse counts when the robot is turned off. This system is calibrated when it is turned on.

A/D value

An analog to digital-value. Converts a multilevel analog electrical system pattern into a digital bit.

AI

Analog input.

AO

Analog output.

alarm

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

algorithm

A fixed step-by-step procedure for accomplishing a given result.

alphanumeric

Data that are both alphabetical and numeric.

AMPS

Amperage amount.

analog

The representation of numerical quantities by measurable quantities such as length, voltage or resistance. Also refers to analog type I/O blocks and distinguishes them from discrete I/O blocks. Numerical data that can vary continuously, for example, voltage levels that can vary within the range of -10 to +10 volts.

AND

An operation that places two contacts or groups of contacts in series. All contacts in series control the resulting status and also mathematical operator.

ANSI

American National Standard Institute, the U.S. government organization with responsibility for the development and announcement of technical data standards.

APC

See absolute pulse code system.

APC motor

See servomotor.

application program

The set of instructions that defines the specific intended tasks of robots and robot systems to make them reprogrammable and multifunctional. You can initiate and change these programs.

arm

A robot component consisting of an interconnecting set of links and powered joints that move and support the wrist socket and end effector.

articulated arm

A robot arm constructed to simulate the human arm, consisting of a series of rotary motions and joints, each powered by a motor.

ASCII

Abbreviation for American Standard Code for Information Interchange. An 8-level code (7 bits plus 1 parity bit) commonly used for the exchange of data.

automatic mode

The robot state in which automatic operation can be initiated.

automatic operation

The time during which robots are performing programmed tasks through unattended program execution.

axis

1. A straight line about which a robot joint rotates or moves. 2. One of the reference lines or a coordinate system. 3. A single joint on the robot arm.

B

backplane

A group of connectors mounted at the back of a controller rack to which printed circuit boards are mated.

BAR

A unit of pressure equal to 100,000 pascals.

barrier

A means of physically separating persons from the restricted work envelope; any physical boundary to a hazard or electrical device/component.

battery low alarm

A programmable value (in engineering units) against which the analog input signal automatically is compared on Genius I/O blocks. A fault is indicated if the input value is equal to or less than the low alarm value.

baud

A unit of transmission speed equal to the number of code elements (bits) per second.

big-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

binary

A numbering system that uses only 0 and 1.

bit

Contraction of binary digit. 1. The smallest unit of information in the binary numbering system, represented by a 0 or 1. 2. The smallest division of a programmable controller word.

bps

Bits per second.

buffer

A storage area in the computer where data is held temporarily until the computer can process it.

bus

A channel along which data can be sent.

bus controller

A Genius bus interface board for a programmable controller.

bus scan

One complete communications cycle on the serial bus.

Bus Switching Module

A device that switches a block cluster to one bus or the other of a dual bus.

byte

A sequence of binary digits that can be used to store a value from 0 to 255 and usually operated upon as a unit. Consists of eight bits used to store two numeric or one alpha character.

C

calibration

The process whereby the joint angle of each axis is calculated from a known reference point.

Cartesian coordinate system

A coordinate system whose axes (x, y, and z) are three intersecting perpendicular straight lines. The origin is the intersection of the axes.

Cartesian coordinates

A set of three numbers that defines the location of a point within a rectilinear coordinate system and consisting of three perpendicular axes (x, y, z).

cathode ray tube

A device, like a television set, for displaying information.

central processing unit

The main computer component that is made up of a control section and an arithmetic-logic section. The other basic units of a computer system are input/output units and primary storage.

channel

The device along which data flow between the input/output units of a computer and primary storage.

character

One of a set of elements that can be arranged in ordered groups to express information. Each character has two forms: 1. a man-intelligible form, the graphic, including the decimal digits 0-9, the letters A-Z, punctuation marks, and other formatting and control symbols; 2. a computer intelligible form, the code, consisting of a group of binary digits (bits).

circular

A MOTYPE option in which the robot tool center point moves in an arc defined by three points. These points can be positions or path nodes.

clear

To replace information in a storage unit by zero (or blank, in some machines).

closed loop

A control system that uses feedback. An open loop control system does not use feedback.

C-MOS RAM

Complementary metal-oxide semiconductor random-access memory. A read/write memory in which the basic memory cell is a pair of MOS (metal-oxide semiconductor) transistors. It is an implementation of S-RAM that has very low power consumption, but might be less dense than other S-RAM implementations.

coaxial cable

A transmission line in which one conductor is centered inside and insulated from an outer metal tube that serves as the second conductor. Also known as coax, coaxial line, coaxial transmission line, concentric cable, concentric line, concentric transmission line.

component

An inclusive term used to identify a raw material, ingredient, part or subassembly that goes into a higher level of assembly, compound or other item.

computer

A device capable of accepting information, applying prescribed processes to the information, and supplying the results of these processes.

configuration

The joint positions of a robot and turn number of wrist that describe the robot at a specified position. Configuration is designated by a STRING value and is included in positional data.

continuous path

A trajectory control system that enables the robot arm to move at a constant tip velocity through a series of predefined locations. A rounding effect of the path is required as the tip tries to pass through these locations.

controller memory

A medium in which data are retained. Primary storage refers to the internal area where the data and program instructions are stored for active use, as opposed to auxiliary or external storage (magnetic tape, disk, diskette, and so forth.)

continuous process control

The use of transducers (sensors) to monitor a process and make automatic changes in operations through the design of appropriate feedback control loops. While such devices historically have been mechanical or electromechanical, microcomputers and centralized control is now used, as well.

continuous production

A production system in which the productive equipment is organized and sequenced according to the steps involved to produce the product. Denotes that material flow is continuous during the production process. The routing of the jobs is fixed and set-ups are seldom changed.

controlled stop

A controlled stop controls robot deceleration until it stops. When a safety stop input such as a safety fence signal is opened, the robot decelerates in a controlled manner and then stops. After the robot stops, the Motor Control Contactor opens and drive power is removed.

controller

A hardware unit that contains the power supply, operator controls, control circuitry, and memory that directs the operation and motion of the robot and communications with external devices. See control unit.

control, open-loop

An operation where the computer applies control directly to the process without manual intervention.

control unit

The portion of a computer that directs the automatic operation of the computer, interprets computer instructions, and initiates the proper signals to the other computer circuits to execute instructions.

coordinate system

See Cartesian coordinate system.

CPU

See central processing unit.

CRT

See cathode ray tube.

cps (viscosity)

Centipoises per second.

CRT/KB

Cathode ray tube/keyboard. An optional interface device for the robot system. The CRT/KB is used for some robot operations and for entering programs. It can be a remote device that attaches to the robot via a cable.

cycle

1. A sequence of operations that is repeated regularly. The time it takes for one such sequence to occur. 2. The interval of time during which a system or process, such as seasonal demand or a manufacturing operation, periodically returns to similar initial conditions. 3. The interval of time during which an event or set of events is completed. In production control, a cycle is the length of time between the release of a manufacturing order and shipment to the customer or inventory.

cycle time

1. In industrial engineering, the time between completion of two discrete units of production. 2. In materials management, the length of time from when material enters a production facility until it exits. See throughput.

cursor

An indicator on a teach pendant or CRT display screen at which command entry or editing occurs. The indicator can be a highlighted field or an arrow (> or ^).

cylindrical

Type of work envelope that has two linear major axes and one rotational major axis. Robotic device that has a predominantly cylindrical work envelope due to its design. Typically has fewer than 6 joints and typically has only 1 linear axis.

D

D/A converter

A digital-to-analog converter. A device that transforms digital data into analog data.

D/A value

A digital-to-analog value. Converts a digital bit pattern into a multilevel analog electrical system.

daisy chain

A means of connecting devices (readers, printers, etc.) to a central processor by party-line input/output buses that join these devices by male and female connectors. The last female connector is shorted by a suitable line termination.

daisy chain configuration

A communications link formed by daisy chain connection of twisted pair wire.

data

A collection of facts, numeric and alphabetical characters, or any representation of information that is suitable for communication and processing.

data base

A data file philosophy designed to establish the independence of computer program from data files. Redundancy is minimized and data elements can be added to, or deleted from, the file designs without changing the existing computer programs.

DC

Abbreviation for direct current.

DEADMAN switch

A control switch on the teach pendant that is used to enable servo power. Pressing the DEADMAN switch while the teach pendant is on activates servo power and releases the robot brakes; releasing the switch deactivates servo power and applies the robot brakes.

debugging

The process of detecting, locating and removing mistakes from a computer program, or manufacturing control system. See diagnostic routine.

deceleration tolerance
The specification of the percentage of deceleration that must be completed before a motion is considered finished and another motion can begin.

default
The value, display, function or program automatically selected if you have not specified a choice.

deviation
Usually, the absolute difference between a number and the mean of a set of numbers, or between a forecast value and the actual data.

device
Any type of control hardware, such as an emergency-stop button, selector switch, control pendant, relay, solenoid valve, or sensor.

diagnostic routine
A test program used to detect and identify hardware/software malfunctions in the controller and its associated I/O equipment. See debugging.

diagnostics
Information that permits the identification and evaluation of robot and peripheral device conditions.

digital
A description of any data that is expressed in numerical format. Also, having the states On and Off only.

digital control
The use of a digital computer to perform processing and control tasks in a manner that is more accurate and less expensive than an analog control system.

digital signal
A single point control signal sent to or from the controller. The signal represents one of two states: ON (TRUE, 1. or OFF (FALSE, 0).

directory
A listing of the files stored on a device.

discrete
Consisting of individual, distinct entities such as bits, characters, circuits, or circuit components. Also refers to ON/OFF type I/O blocks.

disk
A secondary memory device in which information is stored on a magnetically sensitive, rotating disk.

disk memory
A non-programmable, bulk-storage, random-access memory consisting of a magnetized coating on one or both sides of a rotating thin circular plate.

drive power

The energy source or sources for the robot servomotors that produce motion.

DRAM

Dynamic Random Access Memory. A read/write memory in which the basic memory cell is a capacitor. DRAM (or D-RAM) tends to have a higher density than SRAM (or S-RAM). Due to the support circuitry required, and power consumption needs, it is generally impractical to use. A battery can be used to retain the content upon loss of power.

E

edit

1. A software mode that allows creation or alteration of a program. 2. To modify the form or format of data, for example, to insert or delete characters.

emergency stop

The operation of a circuit using hardware-based components that overrides all other robot controls, removes drive power from the actuators, and causes all moving parts of to stop. The operator panel and teach pendant are each equipped with EMERGENCY STOP buttons.

enabling device

A manually operated device that, when continuously activated, permits motion. Releasing the device stops the motion of the robot and associated equipment that might present a hazard.

encoder

1. A device within the robot that sends the controller information about where the robot is. 2. A transducer used to convert position data into electrical signals. The robot system uses an incremental optical encoder to provide position feedback for each joint. Velocity data is computed from the encoder signals and used as an additional feedback signal to assure servo stability.

end effector

An accessory device or tool specifically designed for attachment to the robot wrist or tool mounting plate to enable the robot to perform its intended tasks. Examples include gripper, spot weld gun, arc weld gun, spray paint gun, etc.

end-of-arm tooling

Any of a number of tools, such as welding guns, torches, bells, paint spraying devices, attached to the faceplate of the robot wrist. Also called end effector or EOAT.

engineering units

Units of measure as applied to a process variable, for example, psi, Degrees F., etc.

envelope, maximum

The volume of space encompassing the maximum designed movements of all robot parts including the end effector, workpiece, and attachments.

EOAT

See end of arm tooling, tool.

EPROM

Erasable Programmable Read Only Memory. Semiconductor memory that can be erased and reprogrammed. A non-volatile storage memory.

error

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

error message

A numbered message, displayed on the CRT/KB and teach pendant, that indicates a system problem or warns of a potential problem.

Ethernet

A Local Area Network (LAN) bus-oriented, hardware technology that is used to connect computers, printers, terminal concentrators (servers), and many other devices together. It consists of a master cable and connection devices at each machine on the cable that allow the various devices to "talk" to each other. Software that can access the Ethernet and cooperate with machines connected to the cable is necessary. Ethernets come in varieties such as baseband and broadband and can run on different media, such as coax, twisted pair and fiber. Ethernet is a trademark of Xerox Corporation.

execute

To perform a specific operation, such as one that would be accomplished through processing one statement or command, a series of statements or commands, or a complete program or command procedure.

extended axis

An optional, servo-controlled axis that provides extended reach capability for a robot, including in-booth rail, single- or double-link arm, also used to control motion of positioning devices.

F

faceplate

The tool mounting plate of the robot.

feedback

1. The signal or data fed back to a commanding unit from a controlled machine or process to denote its response to the command signal. The signal representing the difference between actual response and desired response that is used by the commanding unit to improve performance of the controlled machine or process. 2. The flow of information back into the control system so that actual performance can be compared with planned performance, for instance in a servo system.

field

A specified area of a record used for a particular category of data. 2. A group of related items that occupy the same space on a CRT/KB screen or teach pendant LCD screen. Field name is the name of the field; field items are the members of the group.

field devices

User-supplied devices that provide information to the PLC (inputs: push buttons, limit switches, relay contacts, an so forth) or perform PLC tasks (outputs: motor starters, solenoids, indicator lights, and so forth.)

file

1. An organized collection of records that can be stored or retrieved by name. 2. The storage device on which these records are kept, such as bubble memory or disk.

filter

A device to suppress interference that would appear as noise.

Flash File Storage

A portion of FROM memory that functions as a separate storage device. Any file can be stored on the FROM disk.

Flash ROM

Flash Read Only Memory. Flash ROM is not battery-backed memory but it is non-volatile. All data in Flash ROM is saved even after you turn off and turn on the robot.

flow chart

A systems analysis tool to graphically show a procedure in which symbols are used to represent operations, data, flow, and equipment. See block diagram, process chart.

flow control

A specific production control system that is based primarily on setting production rates and feeding work into production to meet the planned rates, then following it through production to make sure that it is moving. This concept is most successful in repetitive production.

format

To set up or prepare a memory card or floppy disk (not supported with version 7.20 and later) so it can be used to store data in a specific system.

FR

See Flash ROM.

F-ROM

See Flash ROM.

FROM disk

See Flash ROM.

G

general override stat

A percentage value that governs the maximum robot jog speed and program run speed.

Genius I/O bus

The serial bus that provides communications between blocks, controllers, and other devices in the system especially with respect to GE FANUC Genius I/O.

gripper

The "hand" of a robot that picks up, holds and releases the part or object being handled. Sometimes referred to as a manipulator. See EOAT, tool.

group signal

An input/output signal that has a variable number of digital signals, recognized and taken as a group.

gun

See applicator.

H

Hand Model.

Used in Interference Checking, the Hand Model is the set of virtual model elements (spheres and cylinders) that are used to represent the location and shape of the end of arm tooling with respect to the robot's faceplate.

hardware

1. In data processing, the mechanical, magnetic, electrical and electronic devices of which a computer, controller, robot, or panel is built. 2. In manufacturing, relatively standard items such as nuts, bolts, washers, clips, and so forth.

hard-wire

To connect electric components with solid metallic wires.

hard-wired

1. Having a fixed wired program or control system built in by the manufacturer and not subject to change by programming. 2. Interconnection of electrical and electronic devices directly through physical wiring.

hazardous motion

Unintended or unexpected robot motion that can cause injury.

hexadecimal

A numbering system having 16 as the base and represented by the digits 0 through 9, and A through F.

hold

A smoothly decelerated stopping of all robot movement and a pause of program execution. Power is maintained on the robot and program execution generally can be resumed from a hold.

HTML.

Hypertext Markup Language. A markup language that is used to create hypertext and hypermedia documents incorporating text, graphics, sound, video, and hyperlinks.

http.

Hypertext transfer protocol. The protocol used to transfer HTML files between web servers.

I

impedance

A measure of the total opposition to current flow in an electrical circuit.

incremental encoder system

A positional information system for servomotors that requires calibrating the robot by moving it to a known reference position (indicated by limit switches) each time the robot is turned on or calibration is lost due to an error condition.

index

An integer used to specify the location of information within a table or program.

index register

A memory device containing an index.

industrial robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions in order to perform a variety of tasks.

industrial robot system

A system that includes industrial robots, end effectors, any equipment devices and sensors required for the robot to perform its tasks, as well as communication interfaces for interlocking, sequencing, or monitoring the robot.

information

The meaning derived from data that have been arranged and displayed in a way that they relate to that which is already known. See data.

initialize

1. Setting all variable areas of a computer program or routine to their desired initial status, generally done the first time the code is executed during each run. 2. A program or hardware circuit that returns a program a system, or hardware device to an original state. See startup, initial.

input

The data supplied from an external device to a computer for processing. The device used to accomplish this transfer of data.

input device

A device such as a terminal keyboard that, through mechanical or electrical action, converts data from the form in which it has been received into electronic signals that can be interpreted by the CPU or programmable controller. Examples are limit switches, push buttons, pressure switches, digital encoders, and analog devices.

input processing time

The time required for input data to reach the microprocessor.

input/output

Information or signals transferred between devices, discreet electrical signals for external control.

input/output control

A technique for controlling capacity where the actual output from a work center is compared with the planned output developed by CRP. The input is also monitored to see if it corresponds with plans so that work centers will not be expected to generate output when jobs are not available to work on.

integrated circuit

A solid-state micro-circuit contained entirely within a chip of semiconductor material, generally silicon. Also called chip.

interactive

Refers to applications where you communicate with a computer program via a terminal by entering data and receiving responses from the computer.

interface

1. A concept that involves the specifications of the inter-connection between two equipments having different functions. 2. To connects a PLC with the application device, communications channel, and peripherals through various modules and cables. 3. The method or equipment used to communicate between devices.

interference zone

An area that falls within the work envelope of a robot, in which there is the potential for the robot motion to coincide with the motion of another robot or machine, and for a collision to occur.

interlock

An arrangement whereby the operation of one control or mechanism brings about, or prevents, the operations of another.

interrupt

A break in the normal flow of a system or program that occurs in a way that the flow can be resumed from that point at a later time. Interrupts are initiated by two types of signals: 1. signals originating within the computer system to synchronize the operation of the computer system with the outside world; 2. signals originating exterior to the computer system to synchronize the operation of the computer system with the outside world.

I/O

Abbreviation for input/output or input/output control.

I/O block

A microprocessor-based, configurable, rugged solid state device to which field I/O devices are attached.

I/O electrical isolation

A method of separating field wiring from logic level circuitry. This is typically done through optical isolation devices.

I/O module

A printed circuit assembly that is the interface between user devices and the Series Six PLC.

I/O scan

A method by which the CPU monitors all inputs and controls all outputs within a prescribed time. A period during which each device on the bus is given a turn to send information and listen to all of the broadcast data on the bus.

ISO

The International Standards Organization that establishes the ISO interface standards.

isolation

1. The ability of a logic circuit having more than one inputs to ensure that each input signal is not affected by any of the others. 2. A method of separating field wiring circuitry from logic level circuitry, typically done optically.

item

1. A category displayed on the teach pendant on a menu. 2. A set of adjacent digits, bits, or characters that is treated as a unit and conveys a single unit of information. 3. Any unique manufactured or purchased part or assembly: end product, assembly, subassembly, component, or raw material.

J

jog coordinate systems

Coordinate systems that help you to move the robot more effectively for a specific application. These systems include JOINT, WORLD, TOOL, and USER.

JOG FRAME

A jog coordinate system you define to make the robot jog the best way possible for a specific application. This can be different from world coordinate frame.

jogging

Pressing special keys on the teach pendant to move the robot.

jog speed

Is a percentage of the maximum speed at which you can jog the robot.

joint

1. A single axis of rotation. There are up to six joints in a robot arm (P-155 swing arm has 8). 2. A jog coordinate system in which one axis is moved at a time.

JOINT

A motion type in which the robot moves the appropriate combination of axes independently to reach a point most efficiently. (Point to point, non-linear motion).

joint interpolated motion

A method of coordinating the movement of the joints so all joints arrive at the desired location at the same time. This method of servo control produces a predictable path regardless of speed and results in the fastest cycle time for a particular move. Also called joint motion.

K

K

Abbreviation for kilo, or exactly 1024 in computer jargon. Related to 1024 words of memory.

KAREL

The programming language developed for robots by the FANUC Robotics America, Inc.

L

label

An ordered set of characters used to symbolically identify an instruction, a program, a quantity, or a data area.

LCD

See liquid crystal display.

lead time

The span of time needed to perform an activity. In the production and inventory control context, this activity is normally the procurement of materials and/or products either from an outside supplier or from one’s own manufacturing facility. Components of lead time can include order preparation time, queue time, move or transportation time, receiving and inspection time.

LED

See Light Emitting Diode.

LED display

An alphanumeric display that consists of an array of LEDs.

Light Emitting Diode

A solid-state device that lights to indicate a signal on electronic equipment.

limiting device

A device that restricts the work envelope by stopping or causing to stop all robot motion and that is independent of the control program and the application programs.

limit switch

A switch that is actuated by some part or motion of a machine or equipment to alter the electrical circuit associated with it. It can be used for position detection.

linear

A motion type in which the appropriate combination of axes move in order to move the robot TCP in a straight line while maintaining tool center point orientation.

liquid crystal display

A digital display on the teach pendant that consists of two sheets of glass separated by a sealed-in, normally transparent, liquid crystal material. Abbreviated LCD.

little-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

load

1. The weight (force) applied to the end of the robot arm. 2. A device intentionally placed in a circuit or connected to a machine or apparatus to absorb power and convert it into the desired useful form. 3. To copy programs or data into memory storage.

location

1. A storage position in memory uniquely specified by an address. 2. The coordinates of an object used in describing its x, y, and z position in a Cartesian coordinate system.

lockout/tagout

The placement of a lock and/or tag on the energy isolating device (power disconnecting device) in the off or open position. This indicates that the energy isolating device or the equipment being controlled will not be operated until the lock/tag is removed.

log

A record of values and/or action for a given function.

logic

A fixed set of responses (outputs) to various external conditions (inputs). Also referred to as the program.

loop

The repeated execution of a series of instructions for a fixed number of times, or until interrupted by the operator.

M

mA

See milliampere.

machine language

A language written in a series of bits that are understandable by, and therefore instruct, a computer. This is a "first level" computer language, as compared to a "second level" assembly language, or a "third level" compiler language.

machine lock

A test run option that allows the operator to run a program without having the robot move.

macro

A source language instruction from which many machine-language instructions can be generated.

magnetic disk

A metal or plastic floppy disk (not supported on version 7.10 and later) that looks like a phonograph record whose surface can store data in the form of magnetized spots.

magnetic disk storage

A storage device or system consisting of magnetically coated metal disks.

magnetic tape

Plastic tape, like that used in tape recorder, on which data is stored in the form of magnetized spots.

maintenance

Keeping the robots and system in their proper operating condition.

MC

See memory card.

mechanical unit

The robot arm, including auxiliary axis, and hood/deck and door openers.

medium

plural **media** . The physical substance upon which data is recorded, such as a memory card (or floppy disk which is not supported on version 7.10 and later).

memory

A device or media used to store information in a form that can be retrieved and is understood by the computer or controller hardware. Memory on the controller includes C-MOS RAM, Flash ROM and D-RAM.

memory card

A C-MOS RAM memory card or a flash disk-based PC card.

menu

A list of options displayed on the teach pendant screen.

message

A group of words, variable in length, transporting an item of information.

microprocessor

A single integrated circuit that contains the arithmetic, logic, register, control and memory elements of a computer.

microsecond

One millionth (0.000001) of a second

milliampere

One one-thousandth of an ampere. Abbreviated mA.

millisecond

One thousandth of a second. Abbreviated msec.

module

A distinct and identifiable unit of computer program for such purposes as compiling, loading, and linkage editing. It is eventually combined with other units to form a complete program.

motion type

A feature that allows you to select how you want the robot to move from one point to the next. MOTYPES include joint, linear, and circular.

mode

1. One of several alternative conditions or methods of operation of a device. 2. The most common or frequent value in a group of values.

N

network

1. The interconnection of a number of devices by data communication facilities. "Local networking" is the communications network internal to a robot. "Global networking" is the ability to provide communications connections outside of the robot's internal system. 2. Connection of geographically separated computers and/or terminals over communications lines. The control of transmission is managed by a standard protocol.

non-volatile memory

Memory capable of retaining its stored information when power is turned off.

O

Obstacle Model.

Used in Interference Checking, the Obstacle Model is the set of virtual model elements (spheres, cylinders, and planes) that are used to represent the shape and the location of a given obstacle in space.

off-line

Equipment or devices that are not directly connected to a communications line.

off-line operations

Data processing operations that are handled outside of the regular computer program. For example, the computer might generate a report off-line while the computer was doing another job.

off-line programming

The development of programs on a computer system that is independent of the "on-board" control of the robot. The resulting programs can be copied into the robot controller memory.

offset

The count value output from a A/D converter resulting from a zero input analog voltage. Used to correct subsequent non-zero measurements also incremental position or frame adjustment value.

on-line

A term to describe equipment or devices that are connected to the communications line.

on-line processing

A data processing approach where transactions are entered into the computer directly, as they occur.

operating system

Lowest level system monitor program.

operating work envelope

The portion of the restricted work envelope that is actually used by the robot while it is performing its programmed motion. This includes the maximum the end-effector, the workpiece, and the robot itself.

operator

A person designated to start, monitor, and stop the intended productive operation of a robot or robot system.

operator box

A control panel that is separate from the robot and is designed as part of the robot system. It consists of the buttons, switches, and indicator lights needed to operate the system.

operator panel

A control panel designed as part of the robot system and consisting of the buttons, switches, and indicator lights needed to operate the system.

optional features

Additional capabilities available at a cost above the base price.

OR

An operation that places two contacts or groups of contacts in parallel. Any of the contacts can control the resultant status, also a mathematical operation.

orientation

The attitude of an object in space. Commonly described by three angles: rotation about x (w), rotation about y (p), and rotation about z (r).

origin

The point in a Cartesian coordinate system where axes intersect; the reference point that defines the location of a frame.

OT

See overtravel.

output

Information that is transferred from the CPU for control of external devices or processes.

output device

A device, such as starter motors, solenoids, that receive data from the programmable controller.

output module

An I/O module that converts logic levels within the CPU to a usable output signal for controlling a machine or process .

outputs

Signals, typically on or off, that controls external devices based upon commands from the CPU.

override

See general override.

overtravel

A condition that occurs when the motion of a robot axis exceeds its prescribed limits.

overwrite

To replace the contents of one file with the contents of another file when copying.

P

parity

The anticipated state, odd or even, of a set of binary digits.

parity bit

A binary digit added to an array of bits to make the sum of all bits always odd or always even.

parity check

A check that tests whether the number of ones (or zeros) in an array of binary digits is odd or even.

parity error

A condition that occurs when a computed parity check does not agree with the parity bit.

part

A material item that is used as a component and is not an assembly or subassembly.

pascal

A unit of pressure in the meter-kilogram-second system equivalent to one newton per square meter.

path

1. A variable type available in the KAREL system that consists of a list of positions. Each node includes positional information and associated data. 2. The trajectory followed by the TCP in a move.

PCB

See printed circuit board.

PC Interface

The PC Interface software uses Ethernet connections to provide file transfer protocol (FTP) functions, PC send macros, telnet interface, TCP/IP interface web server functions, and host communications.

pendant

See teach pendant.

PLC

See programmable logic controller or cell controller.

PMC

The programmable machine controller (PMC) functions provide a ladder logic programming environment to create PMC functions. This provides the capability to use the robot I/O system to run PLC programs in the background of normal robot operations. This function can be used to control bulk supply systems, fixed automation that is part of the robot workcell, or other devices that would normally require basic PLC controls.

printed circuit board

A flat board whose front contains slots for integrated circuit chips and connections for a variety of electronic components, and whose back is printed with electrically conductive pathways between the components.

production mode

See automatic mode.

program

1. A plan for the solution of a problem. A complete program includes plans for the transcription of data, coding for the computer, and plans for the absorption of the results into the system. 2. A sequence of instructions to be executed by the computer or controller to control a robot/robot system. 3. To furnish a computer with a code of instructions. 4. To teach a robot system a specific set of movements and instructions to do a task.

programmable controller

See programmable logic controller or cell controller.

programmable logic controller

A solid-state industrial control device that receives inputs from user-supplied control devices, such as switches and sensors, implements them in a precise pattern determined by ladder diagram-based programs stored in the user memory, and provides outputs for control of processes or user-supplied devices such as relays and motor starters.

Program ToolBox

The Program ToolBox software provides programming utilities such as mirror image and flip wrist editing capabilities.

protocol

A set of hardware and software interfaces in a terminal or computer that allows it to transmit over a communications network, and that collectively forms a communications language.

psi

Pounds per square inch.

Q

queue.

1. Waiting lines resulting from temporary delays in providing service. 2. The amount of time a job waits at a work center before set-up or work is performed on the job. See also job queue.

R

RAM

See Random Access Memory.

random access

A term that describes files that do not have to be searched sequentially to find a particular record but can be addressed directly.

Random Access Memory

1. Volatile, solid-state memory used for storage of programs and locations; battery backup is required. 2. The working memory of the controller. Programs and variable data must be loaded into RAM before the program can execute or the data can be accessed by the program.

range

1. A characterization of a variable or function. All the values that a function can possess. 2. In statistics, the spread in a series of observations. 3. A programmable voltage or current spectrum of values to which input or output analog signals can be limited.

RI

Robot input.

RO

Robot output.

read

To copy, usually from one form of storage to another, particularly from external or secondary storage to internal storage. To sense the meaning of arrangements of hardware. To sense the presence of information on a recording medium.

Read Only Memory

A digital memory containing a fixed pattern of bits that you cannot alter.

record

To store the current set or sets of information on a storage device.

recovery

The restoration of normal processing after a hardware or software malfunction through detailed procedures for file backup, file restoration, and transaction logging.

register

1. A special section of primary storage in a computer where data is held while it is being worked on.
2. A memory device capable of containing one or more computer bits or words.

remote/local

A device connection to a given computer, with remote devices being attached over communications lines and local devices attached directly to a computer channel; in a network, the computer can be a remote device to the CPU controlling the network.

repair

To restore robots and robot systems to operating condition after damage, malfunction, or wear.

repeatability

The closeness of agreement among the number of consecutive movements made by the robot arm to a specific point.

reset

To return a register or storage location to zero or to a specified initial condition.

restricted work envelope

That portion of the work envelope to which a robot is restricted by limiting devices that establish limits that will not be exceeded in the event of any reasonably foreseeable failure of the robot or its controls. The maximum distance the robot can travel after the limited device is actuated defines the restricted work envelope of the robot.

RIA

Robotic Industries Association Subcommittee of the American National Standards Institute, Inc.

robot
A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of tasks.

Robot Model.
Used in Interference Checking, the Robot Model is the set of virtual model elements (sphere and cylinders) that are used to represent the location and shape of the robot arm with respect to the robot's base. Generally, the structure of a six axes robot can be accurately modeled as a series of cylinders and spheres. Each model element represents a link or part of the robot arm.

ROM
See Read Only Memory.

routine
1. A list of coded instructions in a program. 2. A series of computer instructions that performs a specific task and can be executed as often as needed during program execution.

S

saving data.
Storing program data in Flash ROM, to a floppy disk (not supported on version 7.10 and later), or memory card.

scfm
Standard cubic feet per minute.

scratch start
Allows you to enable and disable the automatic recovery function.

sensor
A device that responds to physical stimuli, such as heat, light, sound pressure, magnetism, or motion, and transmits the resulting signal or data for providing a measurement, operating a control or both. Also a device that is used to measure or adjust differences in voltage in order to control sophisticated machinery dynamically.

serial communication
A method of data transfer within a PLC whereby the bits are handled sequentially rather than simultaneously as in parallel transmission.

serial interface
A method of data transmission that permits transmitting a single bit at a time through a single line. Used where high speed input is not necessary.

Server Side Include (SSI)
A method of calling or "including" code into a web page.

servomotor

An electric motor that is controlled to produce precision motion. Also called a "smart" motor.

SI

System input.

signal

The event, phenomenon, or electrical quantity that conveys information from one point to another.

significant bit

A bit that contributes to the precision of a number. These are counted starting with the bit that contributes the most value, of "most significant bit," and ending with the bit that contributes the least value, or "least significant bit."

slip sheet

A sheet of material placed between certain layers of a unit load. Also known as tier sheet.

SO

System output.

specific gravity

The ratio of a mass of solid or liquid to the mass of an equal volume of water at 45C. You must know the specific gravity of the dispensing material to perform volume signal calibration. The specific gravity of a dispensing material is listed on the MSDS for that material.

SRAM

A read/write memory in which the basic memory cell is a transistor. SRAM (or S-RAM) tends to have a lower density than DRAM. A battery can be used to retain the content upon loss of power.

slpm

Standard liters per minute.

Standard Operator Panel (SOP).

A panel that is made up of buttons, keyswitches, and connector ports.

state

The on or off condition of current to and from and input or output device.

statement

See instruction.

storage device

Any device that can accept, retain, and read back one or more times. The available storage devices are SRAM, Flash ROM (FROM or F-ROM), floppy disks (not available on version 7.10 and later), memory cards, or a USB memory stick.

system variable

An element that stores data used by the controller to indicate such things as robot specifications, application requirements, and the current status of the system.

T

Tare

The difference between the gross weight of an object and its contents, and the object itself. The weight of an object without its contents.

TCP

See tool center point.

teaching

Generating and storing a series of positional data points effected by moving the robot arm through a path of intended motions.

teach mode

1. The mode of operation in which a robot is instructed in its motions, usually by guiding it through these motions using a teach pendant. 2. The generation and storage of positional data. Positional data can be taught using the teach pendant to move the robot through a series of positions and recording those positions for use by an application program.

teach pendant

1. A hand-held device used to instruct a robot, specifying the character and types of motions it is to undertake. Also known as teach box, teach gun. 2. A portable device, consisting of an LCD display and a keypad, that serves as a user interface to the KAREL system and attaches to the operator box or operator panel via a cable. The teach pendant is used for robot operations such as jogging the robot, teaching and recording positions, and testing and debugging programs.

telemetry

The method of transmission of measurements made by an instrument or a sensor to a remote location.

termination type

Feature that controls the blending of robot motion between segments.

tool

A term used loosely to define something mounted on the end of the robot arm, for example, a hand, gripper, or an arc welding torch.

tool center point

1. The location on the end-effector or tool of a robot hand whose position and orientation define the coordinates of the controlled object. 2. Reference point for position control, that is, the point on the tool that is used to teach positions. Abbreviated TCP.

TOOL Frame
The Cartesian coordinate system that has the position of the TCP as its origin to stet. The z-axis of the tool frame indicates the approach vector for the tool.

TP.
See teach pendant.

transducer
A device for converting energy from one form to another.

U

UOP
See user operator panel.

URL
Universal Resource Locator. A standard addressing scheme used to locate or reference files on web servers.

USB memory stick
The controller USB memory stick interface supports a USB 1.1 interface. The USB Organization specifies standards for USB 1.1 and 2.0. Most memory stick devices conform to the USB 2.0 specification for operation and electrical standards. USB 2.0 devices as defined by the USB Specification must be backward compatible with USB 1.1 devices. However, FANUC Robotics does not support any security or encryption features on USB memory sticks. The controller supports most widely-available USB Flash memory sticks from 32MB up to 1GB in size.

USER Frame
The Cartesian coordinate system that you can define for a specific application. The default value of the User Frame is the World Frame. All positional data is recorded relative to User Frame.

User Operator Panel
User-supplied control device used in place of or in parallel with the operator panel or operator box supplied with the controller. Abbreviated UOP .

V

variable
A quantity that can assume any of a given set of values.

variance
The difference between the expected (or planned) and the actual, also statistics definitions.

vision system
A device that collects data and forms an image that can be interpreted by a robot computer to determine the position or to "see" an object.

volatile memory

Memory that will lose the information stored in it if power is removed from the memory circuit device.

W

web server

An application that allows you to access files on the robot using a standard web browser.

warning device

An audible or visible device used to alert personnel to potential safety hazards.

work envelope

The volume of space that encloses the maximum designed reach of the robot manipulator including the end effector, the workpiece, and the robot itself. The work envelope can be reduced or restricted by limiting devices. The maximum distance the robot can travel after the limit device is actuated is considered the basis for defining the restricted work envelope.

write

To deliver data to a medium such as storage.

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